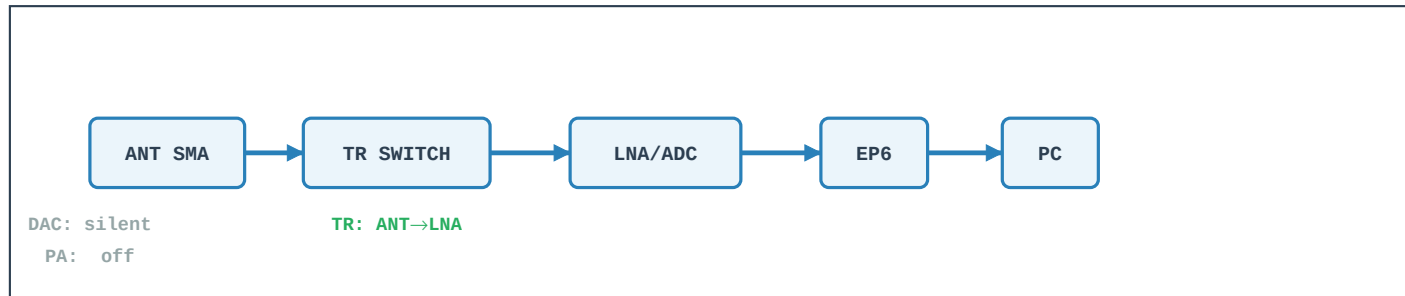


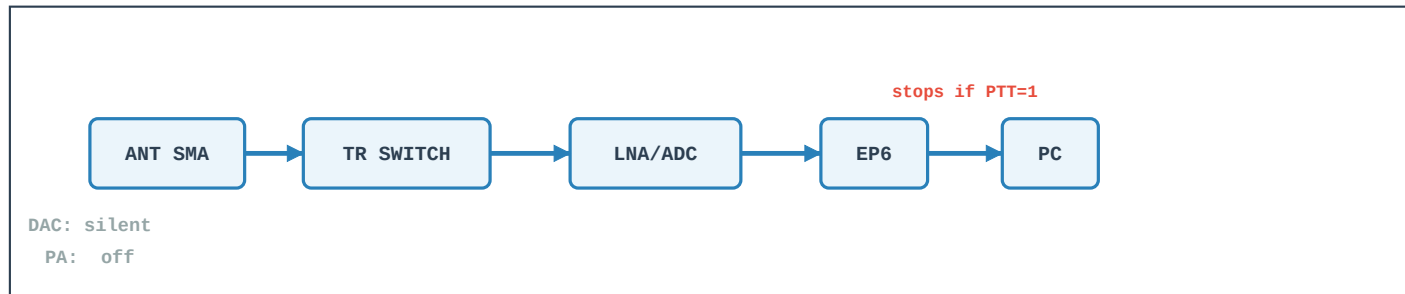
HermesLite 2 – Signal Flow Operating Modes

MODE 1 PTT=0 PA=0 Duplex=1 – Pure Receive



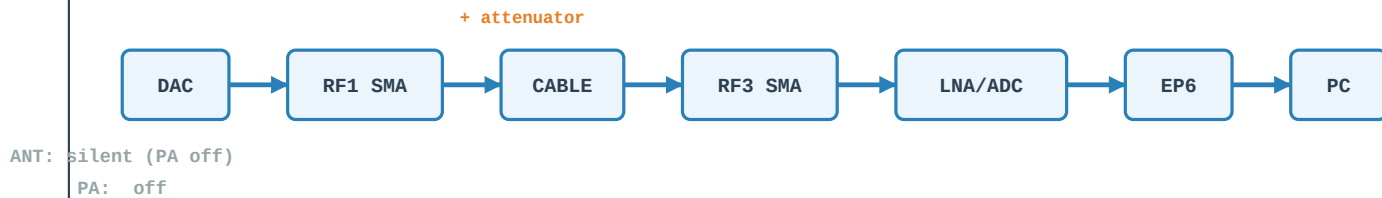
Normal receive. ANT signal enters via TR switch → LNA → ADC. EP6 streams RX data to PC continuously. DAC and PA are silent. TR switch connects ANT port to LNA input.

MODE 2 PTT=0 PA=0 Duplex=0 – Basic Receive Half-Duplex



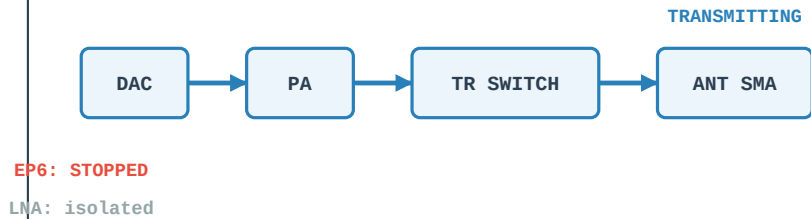
Receive only, half-duplex. Same path as Mode 1 but EP6 streaming stops automatically when PTT goes high. Conventional half-duplex transceiver behaviour – no simultaneous TX/RX.

MODE 3 PTT=1 PA=0 Duplex=1 – RF3 Loopback (current test setup)



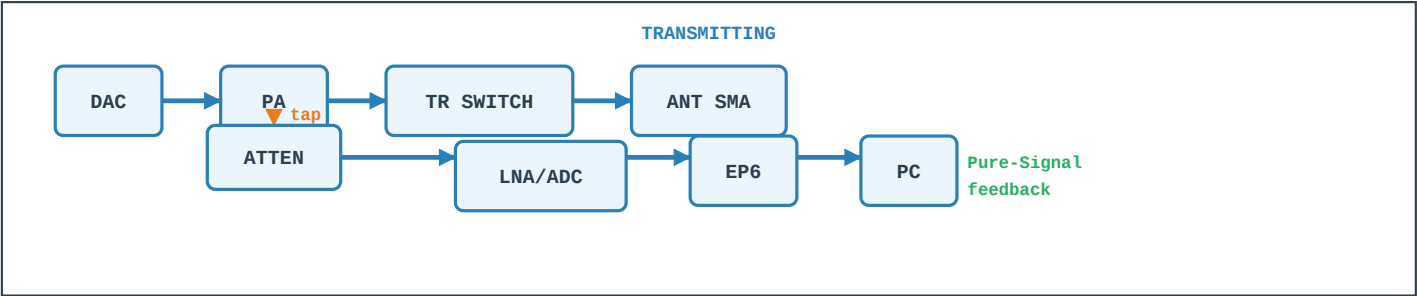
RF3 loopback test mode. DAC drives RF1 SMA; signal routed via external coax + attenuator to RF3 SMA (direct ADC inject). No ANT radiation. PA off. EP6 streams. Clean, calibrated RX test path ideal for SDR calibration and filter testing.

MODE 4 PTT=1 PA=1 Duplex=0 – Transmit Only



Transmit only, half-duplex. DAC → PA → TR switch → ANT. EP6 stops – no RX monitoring during TX. Conventional PTT operation. LNA is protected by TR switch isolation during transmission.

MODE 5 PTT=1 PA=1 Duplex=1 – Transmit + Pure-Signal Feedback



Full transmit with Pure-Signal adaptive predistortion. PA output sampled via fixed tap resistor and attenuator, fed back to ADC. EP6 streams TX feedback to PC during TX. Quisk uses this feedback loop for PA linearisation – significantly reduces IMD products.

Summary Table

Mode	PTT	PA en	Duplex	Signal Route	Use Case
1	0	0	1	ANT→LNA→ADC→PC	Normal receive
2	0	0	0	ANT→LNA→ADC→PC (stops on TX)	Half-duplex receive
3	1	0	1	DAC→RF1→cable→RF3→LNA→ADC→PC	RF3 loopback test
4	1	1	0	DAC→PA→ANT (only)	Transmit, no monitoring
5	1	1	1	DAC→PA→ANT & PA→atten→LNA→ADC→PC	Transmit + Pure-Signal